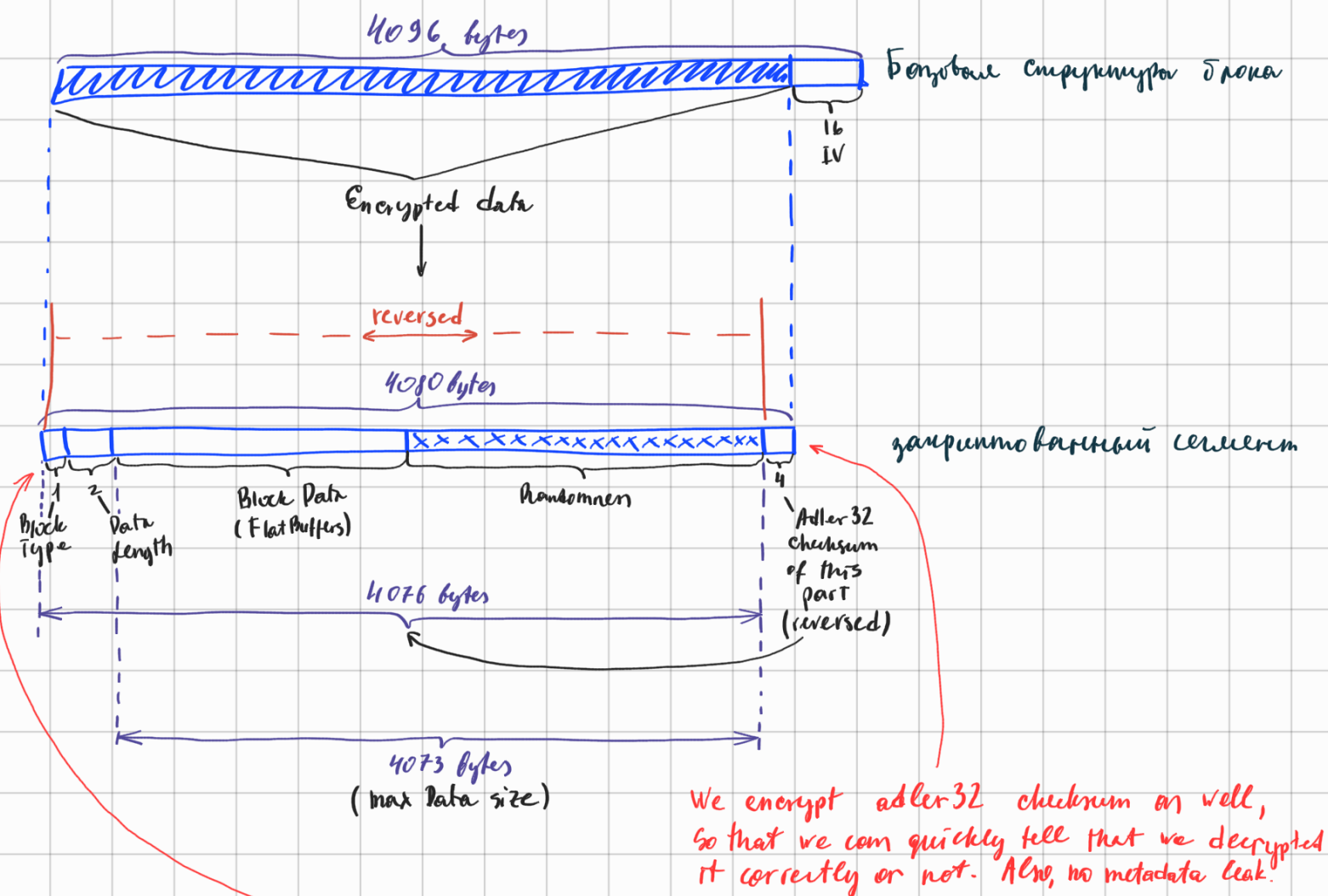
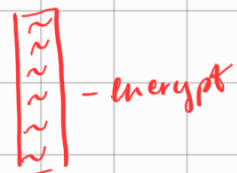
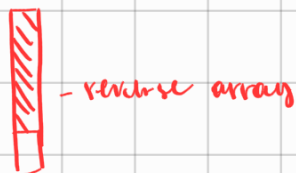




For historical reasons, in Phoner we store Block Version and Tombstoned flag in FlatBuf structure, but conceptually they belong in block header, since they are a part of underlying DB abstraction, which should be agnostic to FlatBuf or Phoner, and reusable for different purposes.  
TO BE FIXED IN FUTURE VERSIONS!!!

- ⑤ - Key Block - this block contains the encryption key to decrypt other blocks
- ③ - Folders - a tree-like structure of folders and subfolders
- ⑤ - Symbol Sets - sets of symbols for word generation (e.g. lower+upper+digits)
- ⑤ - Phrase Templates - templates for phrases (e.g. 'website' → 'url', 'login', 'password')
- Phrases - instances of templates (e.g. if "website": "gmail" → "mail.ru", "user", "pass")  
i.e. actual logins and passwords

- We use AES-256 CBC with IV per block
- Revert Block byte array before encrypting - random bytes go first





## Улучшение секьюрности

### ① Cleanse/Deconstruct ✓ - OK

- Удалить все неактуальные блоки и коллапснуть структуру.

### ② Shuffle ✓ - OK

- Перемешать блоки для предотвращения атаки на KeyBlock, который без этой функции будет находиться в самом начале.

### ③ Triple AES \$ - дорого

- Тут придется писать свою имплементацию AES и это можно под микроконтроллер, это потребует времени, и существенно не изменит факт того, что оно и так секьюрно

|                          |
|--------------------------|
| table Folder             |
| folder-id: uint16        |
| parent-folder-id: uint16 |
| folder-name: string      |

## Folders

|                     |
|---------------------|
| table Folders Block |
| block: StoreBlock   |
| folders: [Folder]   |

|                     |
|---------------------|
| table Key Block     |
| block: StoreBlock   |
| block-count: uint16 |
| key: [byte]         |
| db-name: [byte]     |
| iv: [byte]          |

## encryption key

|                         |
|-------------------------|
| table Symbol Set        |
| set-id: uint16          |
| symbol-set-name: string |
| symbol-set: string      |

## Symbol Sets

|                          |
|--------------------------|
| table Symbol Sets Block  |
| block: StoreBlock        |
| symbol-sets: [SymbolSet] |

|                   |
|-------------------|
| struct StoreBlock |
| block-id: uint16  |
| version: uint32   |
| entropy: uint32   |

|                            |
|----------------------------|
| table Word Template        |
| word-template-id: uint16   |
| permissions: byte          |
| icon: Icon                 |
| min-length: uint16         |
| max-length: uint16         |
| word-template-name: string |
| symbol-set-ids: [uint16]   |

## Phrase Templates

|                                       |
|---------------------------------------|
| table Phrase Template                 |
| phrase-template-id: uint16            |
| phrase-template-name: string          |
| word-template-refs: [WordTemplateRef] |

## Phrases

|                             |
|-----------------------------|
| table Word                  |
| word-template-id: uint16    |
| word-template-ordinal: byte |
| name: string                |
| word: string                |
| permissions: byte           |
| icon: Icon                  |

## Phrase History

|                            |
|----------------------------|
| table Phrase History       |
| phrase-template-id: uint16 |
| phrase: [Word]             |

|                                    |
|------------------------------------|
| table Phrase Template Block        |
| block: StoreBlock                  |
| word-templates: [WordTemplate]     |
| phrase-templates: [PhraseTemplate] |

|                             |
|-----------------------------|
| table WordTemplateRef       |
| word-template-id: uint16    |
| word-template-ordinal: byte |

|                            |
|----------------------------|
| table Phrase Block         |
| block: StoreBlock          |
| phrase-template-id: uint16 |
| folder-id: uint16          |
| is-tombstone: bool         |
| phrase-name: string        |
| history: [PhraseHistory]   |

## Permission flags

- 1 000x - TYPEABLE
- 2 00x0 - VIEWABLE
- 4 0x00 - GENERATEABLE
- 8 x000 - USER-EDITABLE

• enum BlockType; ← not used in structures

• enum Icon;